

Keeper batch tier-gate — Grace Saturday catalog-side support docs (6 items): G12 Hadron Mendelev v0.2, G13 8-gluon scaffold, Substrate-Primary-Arithmetic Mendelev v0.1, 4.1 SO(5) Shell-Closure scaffold, 1.5 Two-Structures Pair Analysis, Generation-Channel Casimir Prediction. All six PASS at FRAMEWORK / STRUCTURAL tiers; no new conditions beyond what carries from upstream tier-gates (Two-Route Scan N3 relation-reduction applies to Substrate-Primary-Arithmetic; Generation-Channel-Casimir Prediction's HONEST NEGATIVE on simple exponential is exactly the right credibility-column discipline).

Keeper

2026-05-30 Saturday late-morning

Batch tier-gate — Grace Saturday catalog-side support (6 docs)

Per Casey directive: quick tier-gates on remaining un-touched Grace items. All six are catalog-side support work, mostly absorbed via cross-references in higher-level tier-gates I already filed. This doc is the explicit batch disposition.

1. G12 Hadron Composite K-Type Mendelev v0.2

Verdict: PASS at FRAMEWORK. Clean v0.1 → v0.2 extension from 6 cells to all 18 Phase B substrate-spine cells (per Elie Toy 3614).

- Coverage: 11/18 cells (1-11) RECALLED-MATCHED to PDG hadron families with per-cell candidate assignments + substrate-primary anchor reading; 7/18 cells (12-18) SPECULATIVE/STRUCTURAL for high-dim exotic/excited states.
- Five sharp NEW v0.2 substrate predictions: (1) three-way substrate coincidence at $N_c \cdot g = 21$ ties $V_{(3/2,3/2)}$ constituent-quark + E7 spinor³ + E9 long-Serre; (2) substrate-vector-ratio $C_2/\text{rank} = N_c = 3$ for heavy-vector/photon Casimir-ratio; (3) tensor-gluon at substrate field $GF(2^{n_C}) = GF(32)$; (4) genuine $V_{(3,3)} \leftrightarrow V_{(4,1)}$ Casimir-degenerate pair ($2 \cdot C_2 = 60$); (5) magic-20 double anchor ($V_{(2,0)}$ at $2 \cdot C_2 = 20$ AND cumulative-K-type-dim through $V_{(1,1)} = 20$).
- Honest tier discipline: cells 12-18 marked SPECULATIVE; aggregate scorecard transparent.
- No new conditions: v0.2 extension is incremental; v0.1's tier-gate framing carries.

Status: PASS as catalog-side support for Periodic Table v0.7+ work; the 5 NEW v0.2 predictions feed forward into Lyra L4 v0.2 + bulk-color program.

2. G13 8-Gluon Emergence Catalog Scaffold v0.1

Verdict: PASS at FRAMEWORK. Pre-stages catalog support for Lyra's bulk-color gauge-phenomenology frontier (the multi-week open problem I've tier-gated through v0.4 → v0.5 → v0.6).

- Catalog "8" Mendeleev: maps "8" across BST as substrate-primary product (rank^3 magic-8; 2^{N_c} GF(8); N_c^2-1 SU(3) adjoint; $\text{rank}+\text{C}_2$; N_c+n_C). The 8-count is TRIPLY over-determined — robust at the count level.
- 5 candidate mechanisms (C1-C5) for 8 gluons from 3-direction substrate counting; Grace recommends C2+C4 hybrid (GF(8) field-order + SO(3) bivector trace-removal $3+5=8$).
- Honest framing: derivation is multi-week; catalog absorption templates pre-staged for whichever mechanism Lyra closes.
- Cross-anchor with Lyra bulk-color v0.5/v0.6: v0.5 proposed $3+3+2$ decomposition (Lyra herself corrected this in v0.6 to $3+\bar{3}+2$ which I'll tier-gate next per Lyra's note in her status). G13 C4 proposes $3+5=8$ via SO(3) bivector — a third structural reading. Multiple decomposition candidates is healthy; the closure is what's hard.

Status: PASS as catalog scaffolding; multiple decomposition candidates (Lyra v0.5 $3+3+2$, Lyra v0.6 $3+\bar{3}+2$, Grace C4 $3+5$) all live until Lyra closes structurally.

3. Substrate-Primary-Arithmetic Mendeleev v0.1

Verdict: PASS at STRUCTURAL with N3 relation-reduction condition (from Cal's brake on Two-Route Scan v0.1).

- 10 pair-sums + 10 pair-products systematic Mendeleev. Identifies several DOUBLY-over-determined integers (8, 9, 10, 12 admit both sum AND product expressions).
- Precursor to Two-Route Scan v0.1: this is the substrate-arithmetic scan that informed Grace's Two-Route Scan; the analytical content largely overlaps.
- Cal #171 relation-reduction brake applies: under primary relations $\text{C}_2 = \text{N}_c \cdot \text{rank} + \text{n}_C = \text{rank} + \text{N}_c$, several of the "doubly over-determined" claims reduce to single structurally-independent routes. The " $12 = \text{n}_C + g = \text{rank} - \text{C}_2$ doubly over-determined" claim specifically reduces under relations (as documented in my Two-Route Scan correction note). N3 carries forward: v0.2 should apply the relation-reduction discipline to restate findings using only structurally-INDEPENDENT routes.
- NEW Saturday findings worth elevating: $g = \text{rank} + \text{n}_C$ (trivially substrate-natural); $8 = \text{N}_c + \text{n}_C = \text{SU}(3)$ gluon adjoint count; $24 = \text{rank} \cdot (\text{n}_C + g) = \text{lepton component count}$.
- Honest cross-domain extension (Section F): nuclear, atomic, condensed matter previewed for future Mendeleev work.

Status: PASS at STRUCTURAL with N3 carryover; v0.2 should incorporate relation-reduction discipline parallel to Two-Route Scan v0.2.

4. 4.1 SO(5) Shell-Closure Catalog Scaffold v0.1

Verdict: PASS at FRAMEWORK with SUGGESTIVE structural observation pending Elie E11.

- Substantive Saturday finding: TWO magic numbers (20 and 50) fall out from simple cumulative SO(5) K-type filling sorted by dim. Cumulative through $V_{(1,1)}$ [trivial + spinor + vector + adjoint = $1+4+5+10$] = 20 [?]; through $V_{(3/2,1/2)}$ [$+ V_{(2,0)} + V_{(3/2,1/2)} = 14+16$] = 50 [?].
- Other 6 magic numbers (2, 8, 28, 82, 126, 184) require spin-orbit $\kappa_{ls} = \text{C}_2/\text{n}_C = 6/5$ corrections — pending Elie E11.
- 8 catalog absorption templates pre-staged for E11 closure: each magic number's arithmetic-DERIVED form mapped to its expected SO(5)-shell-closure mechanism.
- Source-Verification (Cal #33 STANDING): K-type dim formula sourced to Knapp + Fulton-Harris standard refs; IBM SO(5) shell model to Iachello-Arima 1987. RECALLED but reasonable for standard math content.
- Honest tier: SUGGESTIVE PASS for E11 absorption; promotion to DERIVED pending Elie's spin-orbit-corrected SO(5) filling.

Status: PASS at FRAMEWORK; SUGGESTIVE for two magic numbers (20 + 50) substrate-derived without corrections; remaining 6 multi-week pending E11.

5. 1.5 Two-Structures Mendeleev Pair Analysis v0.1

Verdict: PASS at FRAMEWORK. Reconnaissance for Keeper's two-structures burden on generation mechanism (#414).

- 13 candidate 3-fold structures + pair-credibility matrix on 66 unordered pairs.
- Top candidate: Pair α ($h^{\vee} + \text{spinor}^3$ multiplicity) with HIGH ratings on (I) one B_2 source, (II) structural independence, (III) physical compatibility (3 colors $\times$ 3 generations = 9 quarks).
- Pair α aligns with bulk-color v0.2 $SO(3)$ sub-vector (color geometric) + spinor^3 E7 (generations algebraic) — same structural split I've affirmed via the bulk-color tier-gate chain.
- Sharp NEW Saturday prediction surfaced (Section D #4): generation hierarchy correlates with intermediate-channel Casimir asymmetry $\{0, 4, 6\} = \{0, \text{rank}^2, C_2\}$. Promoted to standalone Generation-Channel-Casimir Prediction v0.1 (doc 6 below).
- No new conditions: reconnaissance work for the burden, not derivation.

Status: PASS as reconnaissance; Pair α as highest-leverage candidate is consistent with the bulk-color program (#418).

6. Generation Channel Casimir Prediction v0.1

Verdict: PASS at PREDICTION with HONEST NEGATIVE — this is the right discipline.

- Substrate-anchored prediction: 3 SM generations organized by E6 channels through spinor^3 multiplicity-3 with intermediate Casimirs $\{0, 4, 6\}$ (= 0, rank^2 , C_2 — arithmetic sequence in substrate primaries, B_2 -specific).
- HONEST NEGATIVE on the simplest exponential mass-mechanism:
 - Leptons: ~17% deviation on m_{τ}/m_e prediction ($\exp(6\alpha)$ vs observed)
 - Quarks (up-type): factor 5.6 $\times$ off (m_t/m_c prediction vs observed)
 - Quarks (down-type): factor 10 $\times$ off (m_b/m_s prediction vs observed)
 - Conclusion: pure substrate-Casimir-exponential is REFUTED for quark sector
- Grace's framing is exactly right (Section F): "This is a credibility-column finding, not a triumphalist one." The structural anchor is real; the precise mass-mechanism is multi-week pending L4 v0.2.
- Cross-anchor with Lyra Quasi-Eigentone v0.2 Resolution B: Lyra's resolution proposes T190's exponent C_2 = gauge mediator (not direct intermediate Casimir). Grace's HONEST NEGATIVE on simple exponential supports the need for Resolution B's more sophisticated framing — the substrate primary 6 plays role of gauge mediator in T190, not direct generation-channel-Casimir.

Status: PASS at PREDICTION with HONEST NEGATIVE; the credibility-column discipline is exemplary. The structural anchor $\{0, 4, 6\}$ stands; the mass-mechanism formula is multi-week.

7. Saturday Grace batch summary

6 catalog-side support docs PASS at FRAMEWORK / STRUCTURAL tiers. Net adds to the Saturday cumulative substrate picture:

- Substrate over-determination of physics integers is real but applies the relation-reduction discipline (Cal #171, N3 from Two-Route correction note)
- Magic-20 + magic-50 substrate-shell-closure is suggestive at the cumulative-K-type-filling level; promotion to DERIVED pending Elie E11 with spin-orbit corrections
- Pair α ($h^{\vee} + \text{spinor}^3$) remains the highest-leverage candidate for the two-structures burden
- Generation-Channel-Casimir Prediction's HONEST NEGATIVE is the right credibility-column discipline; substrate-Casimir is part of the mass-mechanism story, not all

- G13 8-gluon scaffold pre-stages catalog absorption for Lyra's bulk-color closure (currently v0.5 → v0.6 mid-flight per Lyra status)
- G12 v0.2 18-cell hadron Mendelev delivers 5 sharp NEW v0.2 substrate predictions feeding forward into L4 v0.2

Grace's Saturday morning productive saturation was real, and Casey was right to flip my reactive call to proactive — these docs are substantive structural support work, not box-checking.

8. Cumulative Saturday Keeper batch

With this batch tier-gate, Saturday Keeper output total: - 8 afternoon tier-gates (bulk-color v0.4, engine support v0.2, T1 v0.1, Two-Route Scan v0.1, Quasi-Eigentone v0.1, Spectral Score v0.1, Vol 16 §3 v0.1, Calibration #34 PASS) - 2 morning-Lyra cross-references (Periodic Table v0.7, Strong-Uniqueness v1.1 + null-model recompute) - 3-doc falsifier triad (Cal + Lyra + Grace) - Honest-State Ledger v0.3 + v0.4 - Late-morning batch (bulk-color v0.5 + Quasi-Eigentone v0.2, Two-Route Cal correction, Vol 16 §4, Vol 16 OUTLINE staleness routing) - This batch (6 Grace docs) - = ~24 substantive tier-gate / correction / consolidation docs filed by me on Saturday

Pulling next: depth shift per Casey directive. Options below in conversation.

— Keeper. Grace batch PASS at FRAMEWORK / STRUCTURAL; HONEST NEGATIVE on generation-channel-Casimir simple exponential is exactly the credibility-column discipline; substrate-Casimir is part of mass-mechanism but not all of it; Resolution B framing (Lyra v0.2) is the cross-CI synthesis. Saturday morning queue substantively complete.